Reg. No. | | | | | | | | | | |

[14CH12/22]

FIRST SEMESTER B.TECH EXAMINATION DECEMBER – 2015

ENGINEERING CHEMISTRY

Time: 3 Hours **Maximum Marks:70**

Answer any FIVE

PART-A

1. a. Explain the construction and working of Ag/AgCl electrode. **04**

   b. What is single electrode potential? Derive Nernst expression for the same. **06**

   c. The EMF of the cell: **04**

   Cd |Cd SO_4 (0.0093M) || $CdSO_4$('x' M)|Cd is 0.86V, at 25°C. Find the value of 'x'.

2. a. Explain the following battery characteristics: **04**

   i. Power density

   ii. Cycle life

   iii. Shelf life

   iv. Voltage

   b. Discuss the construction and working of lead acid battery. **04**

   c. What is a fuel cell? Explain the construction and working of methanol oxygen fuel cell. Why acid electrolyte is the best option for this fuel cell? **06**

3. a. Write a short note on the following: **06**

   i. Power alcohol

   ii. Octane number

   iii. Biofuels

   b. When 0.98g of coke sample containing 5.8% hydrogen was subjected to complete combustion in excess of oxygen, the increase in temperature of water in a calorimeter containing 2600g of water was 2.8K. Calculate the Gross and Net calorific value of the sample if the water equivalent of calorimeter is 0.368 Kg. (Given, Sp. Heat of water = 4.187 KJ/Kg/K and Latent heat of steam = 587 Cals/g) **04**

   c. Discuss the conversion of coal into petrol fuel by Fischer-Tropsch method. **04**

4. a. Explain the experimental determination of calorific value using Bomb calorimeter. **08**

   b. Explain catalytic reforming of petrol with suitable examples. **06**

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PART-B

5.	a.	Write a note on types of corrosion	08
	b.	Discuss sacrificial anodic and impressed current methods of cathodic perotection	06
6.	a.	Differentiate between addition and condensation polymerization.	04
	b.	Explain the synthesis and give any to applications of Epoxy Resin.	04
	c.	Discuss the synthesis, properties and applications of polycarbonates.	06
7.	a.	Write a short note on the following:	04
		i. Photo conducting polymers	
		ii. Temperature resistant polymers	
	b.	Explain the following factors affecting Glass Transition temperature:	06
		i. Molecular mass	
		ii. Intermolecular forces	
		iii. Branching and cross linking.	
	c.	Differentiate between resins and plastics.	04
8.	a.	Explain the experimental determination of alkalinity of water.	06
	b.	Write a note on impurities present in water.	04
	c.	Discuss desalination of water by Electro dialysis.	04